

Quantum Learning

A Concept for a New Educational System

From Educational Programs to Managing Changes in Human Capability

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Abstract

The modern educational system is organized around large, preassembled units: the course, discipline, semester, program, qualification, and degree. This architecture emerged in a world where knowledge changed relatively slowly, professions remained stable, and professional capabilities had long life cycles. Today, technologies and workforce requirements change faster than universities can update their programs. Employers need a continuous reskilling system, while working adults need a precise way to close the gap between their current capabilities and the next task. Learning becomes their third job, alongside paid employment and family responsibilities. At the labor-market level, this gap appears as structural unemployment: vacancies and job seekers coexist, but the graphs of required and available capabilities do not match. Artificial intelligence accelerates this mismatch by automating some classes of tasks while intensifying labor shortages in others. This paper proposes the concept of quantum learning: an event-driven, discrete, continuous, and executable system for managing changes in human capability. Its fundamental unit is the learning quantum—a minimal, completed, and verified transition from one capability state to another. Quantum learning replaces the linear educational program with an individual graph of transitions. Students, teachers, mentors, experts, employers, and educational organizations are represented by bounded digital twins. Tasks, capabilities, quanta, attempts, errors, evidence, and authorizations exist as autonomous actors in an executable environment. Changing the unit of learning also changes the economics of education. A student pays for the individual pathway actually completed, while capabilities already demonstrated reduce its duration and price. The university moves from a one-time sale of a degree to the continuous stewardship of the graphs of people and companies; the teacher scales expertise through reusable quanta, simulations, AI, and work on complex exceptions. Simulator is the Corezoid Inc. platform on which the author is implementing this architecture. The concept itself is technologically neutral and can be implemented in any environment capable of executing an event-driven graph of actors, states, and transitions. The primary learner for whom this system is designed is the working adult, who simultaneously carries three jobs: paid employment, family responsibilities, and continuous learning.

Keywords: quantum learning; capability-based education; learning graphs; digital twins; event-driven systems; workforce reskilling; structural unemployment

A course measures content. A quantum measures change in the person.

Part I. Foundations of Quantum Learning

1. Education Can No Longer Keep Pace with the World

The world changes faster than a person can complete an educational program. Professions are reconfigured faster than universities revise their disciplines. Tools become obsolete faster than employees complete corporate courses. Employers need new capabilities now, while the educational system offers semesters, programs, and diplomas months or years later. Studying for three, four, or five years before beginning to act is too slow. Retraining for months every time a technology, process, or role changes is also too slow.

The problem is not a shortage of teachers, platforms, video lectures, or educational content. The problem lies in the architecture of education itself. It is still built around large, preassembled packages:

- a course
- a discipline
- a semester
- a program
- a degree
- a diploma.

The content is predetermined. The duration is fixed in advance. The sequence of topics is set in advance. Completion of the program is treated as the result, rather than a demonstrated change in what the person can do. This model was created for a world in which knowledge changed relatively slowly, professions lasted for decades, and learning could be separated from work.

That world no longer exists.

1.1. Learning Has Become a Third Job

“A working adult today has not two jobs, but three. The first is paid work. The second is family—children, loved ones, the home, and pets. The third is continuous learning. The world is changing, and this is permanent. Learning therefore cannot be designed as an unpaid add-on to evenings and weekends.” — Alexander Vityaz

An adult’s first job is paid professional work. The second is family: children, loved ones, older relatives, domestic responsibilities, and pets. Continuous learning becomes a third job and competes with the first two for time, attention, energy, and recovery. It cannot be designed as an optional hobby after the working day. When an organization requires continuous capability renewal, it must account for the learning load in working time, funding, role planning, and the evaluation of results. A quantum architecture must reduce the cost of this third job without damaging the first two: it must recognize what has already been demonstrated, remove unnecessary segments of the pathway, connect learning to a real task, support flexible formats, and introduce human assistance only where it is genuinely necessary.

Here, “job” means a persistent circuit of obligations requiring time, attention, planning, and responsibility. The concept is broader than paid employment: family, care, the home, pets, and learning also create recurring demands that compete for an adult’s limited resources.

LEARNERS' LIFE-LOAD MATRIX

How many persistent "jobs" do different categories of learners carry?

■ Paid work / business
 ■ Family / care
 ■ Learning
 ■ Additional work / other obligations
 ■ Home / pets

Category	1st job	2nd job	3rd job	4th job	Total
Traditional student	Study	Part-time job?	—	—	1–2
Single working adult	Work	Learning	Pets? / home	—	2–3
Working adult in a couple	Work	Family / home	Learning	—	3
Adult with children	Work	Family / children	Learning	Pets? / care	3–4
Single parent or caregiver	Work	Children / care	Learning	Home / pets?	3–4
Entrepreneur / manager	Business	Team / family	Learning	Other obligations?	3–4

"Job" means a persistent circuit of obligations requiring time, attention, planning, and responsibility—not only paid employment. Dashed outline = optional load.

Figure 1. Learners' Life-Load Matrix. Adult learning competes not with "empty time," but with existing demands from employment, family, care, domestic work, and recovery. Dashed outlines indicate optional loads.

The matrix does not establish a universal social norm or describe every person literally. It presents typical combinations of responsibilities. For many adults, three jobs are the minimum; single parenthood, caring for relatives, entrepreneurship, domestic work, or pets create a fourth persistent circuit. A university or employer therefore has no right to assume that evenings and weekends constitute free learning capacity.

Participant	Practical implication
Student	Protected time, a clear pathway, recognition of what has already been demonstrated, and the ability to learn without losing income or withdrawing from family life.
Employer	Learning becomes part of the production model rather than an employee's private evening activity: working time, co-funding, real tasks, a mentor, and feedback.
University	Programs give way to flexible individual pathways compatible with work, family, care for loved ones and pets, and recovery.
Teacher	Work is organized around the context of adult students, their limited time, complex interventions, and evidence of transition—not the repetition of the same lecture.
State and regulator	Portable evidence, shared funding, and protection of an adult's right to reskill without a double financial and family penalty.

1.2. Structural Unemployment: Jobs and People Exist, but the Transition Does Not

Artificial intelligence does not merely reduce or increase the total number of jobs. It changes the structure of labor demand. Demand for some office, administrative, and content-related functions may decline, while manufacturing, construction, energy, logistics, maintenance, and other sectors simultaneously face labor shortages. A review published by *Ekonomichna Pravda* on July 22, 2026, describes this process as a deepening of structural unemployment in Ukraine: AI reduces demand for some classes of labor while intensifying shortages in others [2].

This is a structural problem, not merely a cyclical one. Vacancies exist. Job seekers also exist. But the sets of capabilities, work contexts, authorizations, geographic conditions, physical constraints, and terms of employment do not match. The International Labour Organization defines skills mismatch precisely as a discrepancy between the capabilities employers seek and those individuals possess [3]. A copywriter cannot replace a welder. Releasing a person from one occupation does not automatically turn that person into a resource for another industry. Labor is not a homogeneous mass that can be mechanically reassigned from one vacancy to another.

Every vacancy is a graph of tasks, required capabilities, tools, risks, and authorizations. Every person is another graph: demonstrated capabilities, transferable experience, constraints, goals, and possible next transitions. Structural unemployment arises when these graphs do not match and the system cannot compute and execute a transition between them. The traditional educational system responds to such a gap with another long program. But a person whose skills have lost market relevance today, and an employer with an unfilled role today, do not need a generic course. They need the shortest admissible pathway: which capabilities already transfer; which quanta are missing; which constraints make the pathway impossible; which simulations and mentors are

required; and what evidence will authorize access to real work.

In this sense, quantum learning is more than an educational technology. It becomes infrastructure for labor reallocation. The system must match a person's digital twin against the graphs of unfilled work, compute several realistic pathways, estimate their time, cost, and risk, guide the person through the necessary quanta, and verify readiness in real work.

Jobs exist. People exist. What is missing is an executable graph of transitions between them.

STRUCTURAL UNEMPLOYMENT: JOBS AND PEOPLE EXIST, BUT THE TRANSITION DOES NOT

The problem is a mismatch between the person's graph and the graph of work requirements—not an absence of labor.

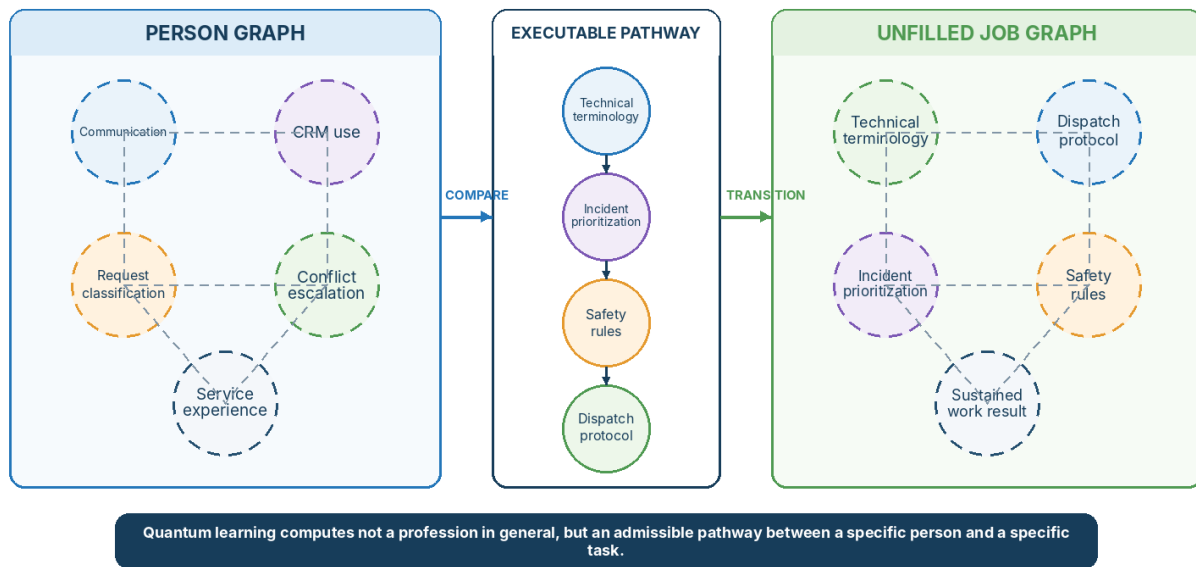


Figure 2. Structural unemployment as a mismatch between the graph of a person's demonstrated capabilities and the graph of requirements for unfilled work. Quantum learning constructs an admissible pathway between them, accounting for time, cost, risk, simulation, and authorization.

2. The Course Is No Longer the Primary Unit of Learning

A course measures the volume of content delivered. But an employer does not need a measure of content volume. The employer needs to know:

- what the person can do
- how independently the person can do it
- under what conditions the capability was demonstrated
- the quality of the result
- when the capability was last verified
- whether it corresponds to the current task
- whether it persists when the context changes.

The fact that a lecture was attended proves nothing. Completing a course does not prove readiness to act. An examination often demonstrates only the ability to reproduce material under artificial conditions. Education must measure not the transmission of knowledge, but the change in a person's capability to act.

The primary unit of the new education is therefore neither the course, the subject, the semester, nor the program. The primary unit is the **learning quantum**.

FROM A LINEAR PROGRAM TO AN INDIVIDUAL LEARNING GRAPH

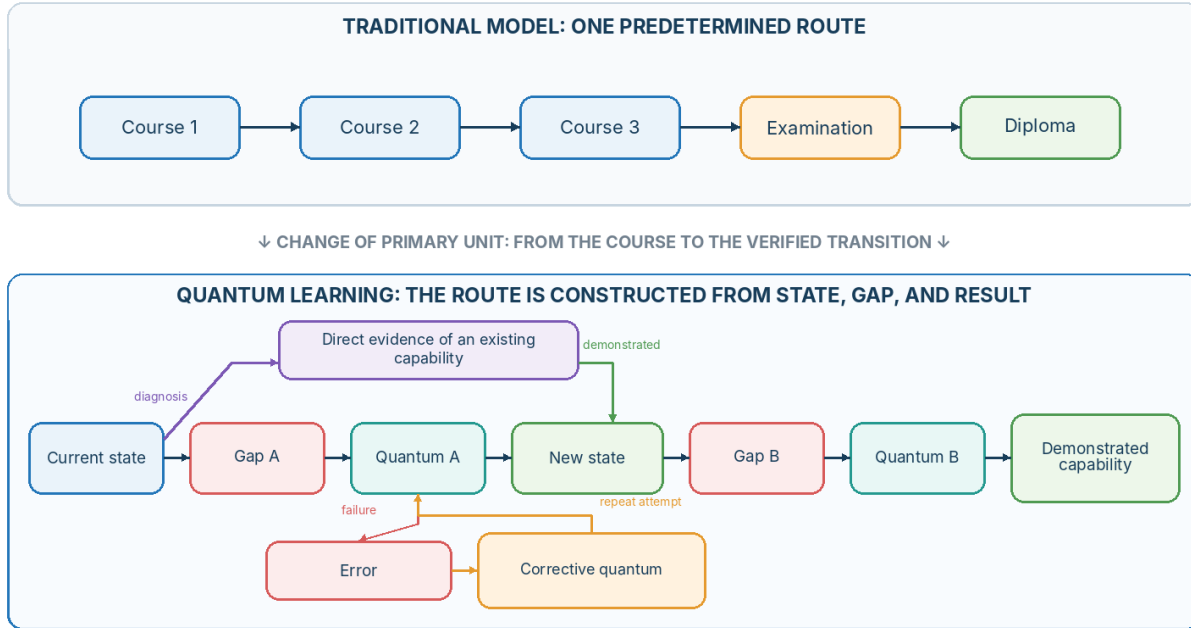


Figure 3. A linear educational program and an individual graph of quantum learning. In the traditional model, the sequence is predetermined. In quantum learning, the pathway is constructed from the current state, identified gaps, attempts, and verified transitions.

3. What Is a Learning Quantum?

A learning quantum is the smallest completed and verifiable transition from one state of capability to another. The treatment of transition as the primary educational unit is conceptually aligned with the author’s broader process ontology, in which transitions rather than static objects form the primitive structure of system description [10]. A person could not perform a specific action. The person received exactly the knowledge, explanation, exercise, simulation, or guidance required to perform that action. The person performed the action. The result was assessed.

The change in capability was demonstrated and recorded.

Could not. Learned. Did. Demonstrated.

That is a learning quantum. The word “quantum” is used here to denote the discreteness of a verified transition, not quantum computing. A short video, a micro-module, or an isolated prompt is not a quantum in itself. Minimality refers not to duration, but to the substantive integrity of the transition. One quantum may take minutes; another may require months of practice, observation of a master practitioner, and a series of demonstrations across different contexts.

Q: $S0 \rightarrow S1$

A quantum is not measured by the number of videos watched, pages read, or hours spent in a classroom. A short video does not become a quantum merely because it is short. A micro-course does not become quantum learning merely because it has been divided into small components. A quantum exists only when a capability transition has occurred and been demonstrated.

Learning has occurred only when what a person is capable of doing has changed. Everything else is information, preparation, or an imitation of the educational process.

MINIMUM STRUCTURE OF A LEARNING QUANTUM

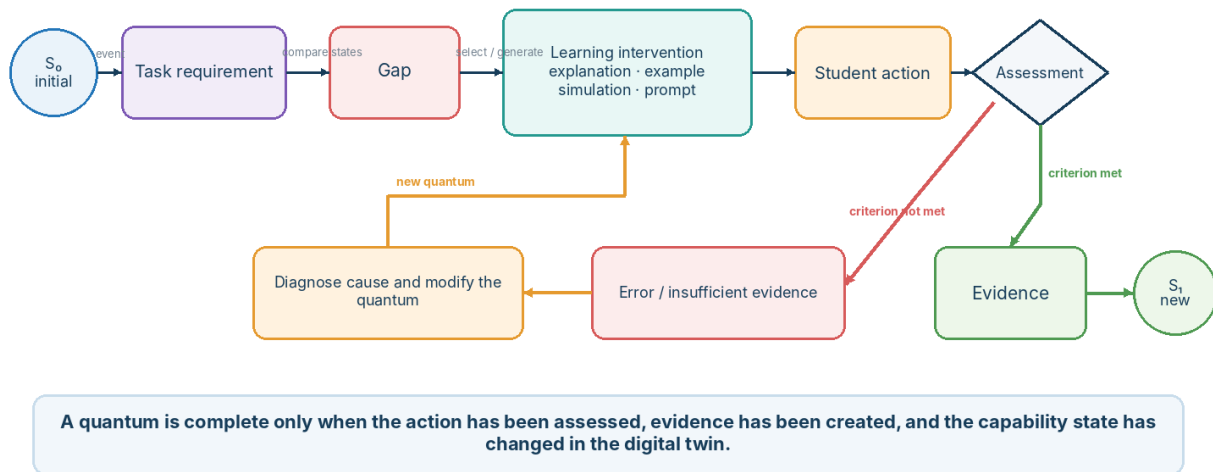


Figure 4. A learning quantum as a completed and verifiable transition. A learning intervention becomes a quantum only after the student acts, the action is assessed, evidence is created, and the capability state changes.

4. Capability as the Primary Object of Learning

In the new architecture, the primary object of management is not content, but capability. A capability is a verified capacity to perform a specific action or class of actions:

- under specified conditions
- at an established level of quality
- with an admissible level of assistance
- with a defined degree of independence
- within a specified risk boundary.

Capability cannot be reduced to knowledge. A person may know the rules of negotiation yet be unable to conduct a real negotiation. A person may know the syntax of a programming language yet be unable to correct a failure in a production system. A person may have studied financial analysis yet be unable to make a management decision on that basis. The statement “the person knows financial modeling” is therefore insufficient.

An operational description of the capability should instead read:

The person can independently construct a cash-flow model from a specified data set, identify a cash shortfall, explain its causes, and test the robustness of the underlying assumptions.

A capability has at least the following dimensions:

- | | | |
|--------------------|-------------------------|--|
| • action | • constraints | • complexity |
| • object of action | • quality of the result | • risk |
| • context | • level of independence | • date of the most recent verification |
| • tools | • admissible assistance | • period of validity. |

4.1. Being Able to Solve Specific Problems: The “Mathematical Trivium”

In 1991, V. I. Arnold published *The Mathematical Trivium*—a list of one hundred relatively simple problems intended as a necessary minimum of practical mathematical qualification. His criterion was operational: a specialist is defined not by the complexity of the terminology they command, but by the problems they can solve with their own hands [6].

*“What is at issue here is not any difficult problem, but the simple questions that constitute the strictly necessary minimum. There need not be many such problems, but the ability to solve them must be required rigorously.” — V. I. Arnold, *The Mathematical Trivium*, 1991*

Quantum learning generalizes this principle beyond mathematics. Every qualification should have its own professional trivium: a compact and explicit set of classes of tasks that a person must be able to perform independently, at a specified level of quality. This is not a complete curriculum, but a strict operational minimum.

A university must answer not only the question “Which topics were studied?” but, above all, “Which specific tasks can this person perform, in what context, with what degree of independence, and on the basis of what evidence?” For a procedural capability, one rigorous practical assessment may be sufficient evidence; for professional judgment, a series of decisions across multiple contexts may be required. A qualification claim without a task and a result remains insufficient.

5. The State of a Capability

A person does not simply “have” or “lack” a capability. A capability has a state.

For example:

Not diagnosed → deficit identified → understands the principle → performs by following an example → performs with a prompt → performs under supervision → performs independently → transfers to a new context → can assess others → reverification required.

A capability state is described not by a single grade, but by a set of parameters: action, context, quality of the result, independence, assistance used, and currency of the evidence.

This model makes it possible to replace vague lists of skills with observable states.

6. Learning Begins with a Gap

Traditional education begins with a program. Quantum learning begins with the gap between what a person is required to do and what that person can do now. Let:

- S_c be the current capability state
- S_r be the state required by the task.

The gap is recorded as a structural difference between the required and current capability states. For each component, the system must specify what is missing: knowledge of a rule, recognition of a situation, quality, independence, speed, transfer to a new context, or the necessary authorization. A gap may arise because a person:

- does not know the rule
- does not understand the context
- does not recognize the situation
- cannot select the appropriate action
- uses an obsolete tool
- can act only with prompting
- does not meet the quality criterion
- cannot transfer the capability to a new context
- makes a critical error
- performs the action too slowly
- is not ready for the required level of risk.

Once the gap has been identified, the system must generate not an entire course, but the minimal transition required to close it.

7. Learning Is Triggered by an Event

Traditional learning is triggered by a schedule. Quantum learning is triggered by an event. An event may be:

- the emergence of a new task
- a change in a work process
- the introduction of a new tool
- a change in legislation
- an employee's transition to another role
- an error
- a decline in quality
- an unsuccessful attempt
- expiration of the capability's validity period
- the emergence of a new professional goal
- a change in employer requirements.

The event creates a new requirement or reveals a mismatch between the task and the person's current state. The system must then:

1. identify the required capability
2. retrieve the current state
3. compute the gap
4. create a gap actor
5. select or generate a learning quantum
6. register the person's action
7. assess the result
8. create evidence
9. change the capability state
10. recompute the subsequent pathway.

Learning ceases to be a separate activity.

It becomes part of the activity itself.

Part II. Digital Twins and Actors in the Educational System

8. Quantum Learning Is Impossible Without Digital Twins

Quantum learning cannot be implemented through a content library, a course management system, or a chatbot alone. Managing a transition requires more than a single student profile. It requires an executable graph of bounded digital twins. The human actors are the student, teacher, mentor, expert, employer, and educational organization. The executable entities are the task, capability, gap, quantum, attempt, error, evidence, simulation, assessment rule, and authorization. AI is also an actor, with explicitly defined powers.

This is not a collection of records. Every actor has a state, receives events, performs actions, creates evidence, and affects the decisions of other actors. The model stores only the data required for a specific transition and its assessment [9].

9. The Conant–Ashby Theorem as an Architectural Foundation

The need for current models of actors is motivated by more than the convenience of automation.

The Conant–Ashby Good Regulator Theorem establishes a general requirement for any system of regulation [1]:

Every good regulator of a system must be a model of that system.

Every participant in the learning process performs a regulatory function. The student regulates their own actions. The teacher regulates the learning intervention. The mentor regulates the transition toward independence.

The expert regulates the verification of a complex capability. The employer regulates requirements and authorization for real work. The educational organization regulates resources, trajectories, and the quality of transitions. AI regulates the local selection of explanations, exercises, and feedback.

Every regulator must possess a sufficiently rich model of the system about which it makes decisions. The teacher must have a model of the student. The student must have a model of the task, the goal, and the criteria for the result. The employer must have a model of the required role and the person's demonstrated readiness.

The educational organization must have models of students, teachers, employers, tasks, quanta, and its own capabilities. AI must receive from Simulator the current states of all actors whose transition it helps organize. Without a model, the regulator decides blindly. With an outdated model, it regulates a past state.

With a model that is too coarse, it cannot distinguish the system's consequential states.

No model of an actor—no good regulation of that actor. No graph of digital twins—no governable educational system.

The theorem does not imply that a particular platform or a single model format is mandatory. The engineering implication follows from the law of requisite variety: when tasks, roles, and human states change faster than an administrator can update static profiles, the model must update in response to events, remain machine-readable, and participate in the execution of decisions.

A digital twin is the proposed technical implementation of this requirement. Its quality is assessed not by the volume of data collected, but by the accuracy of decisions, the speed of adaptation, and the capacity to correct its own errors.

GRAPH OF DIGITAL TWINS IN QUANTUM LEARNING

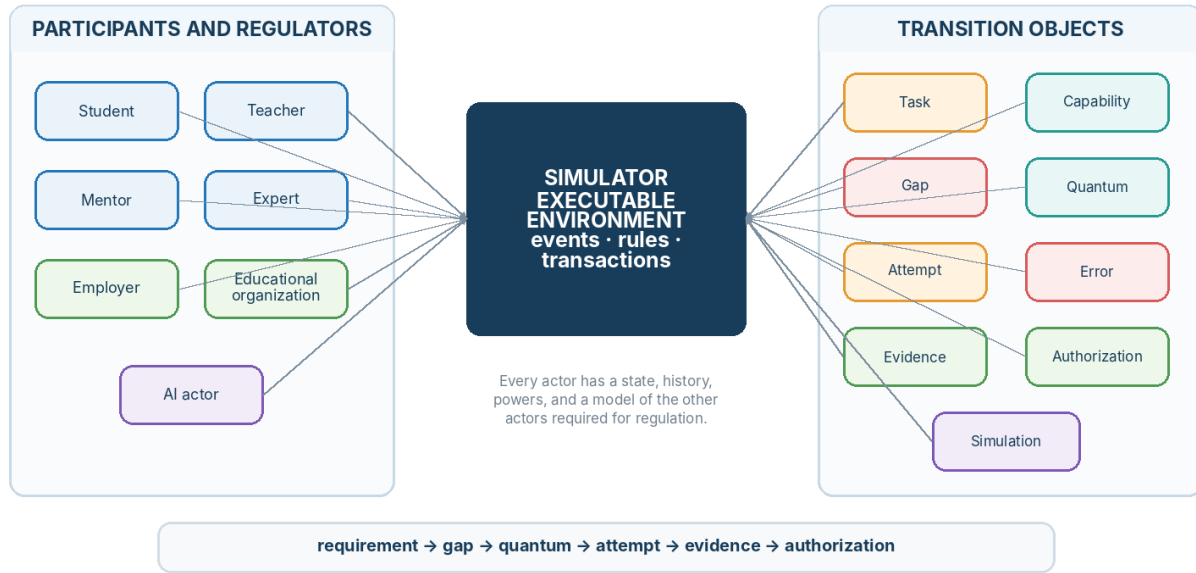


Figure 5. Quantum learning as a graph of digital twins. Every participant and every consequential object in the process is represented as an actor because the state of each affects how the transition is regulated.

9.1. The Boundaries of the Digital Model

A person's digital twin is not a digital copy of their personality. It is a bounded, contextual, and contestable model of the states relevant to a particular educational or professional decision. It does not answer the question "Who is this person?" It answers a narrower question: which actions the person has demonstrably performed, under what conditions and with what assistance, when those actions were verified, and how relevant the evidence is to the current task.

Every claim in the model must have a context, source, confidence level, validity period, and review procedure. The model must not become a universal rating of a person's personality, talent, motivation, or future.

9.2. Adaptation, Development, and Exploration

Quantum learning operates in three distinct modes. In the adaptation circuit, the target action, criterion, and risk are known; the transition can therefore be designed with the greatest rigor. In the development circuit, the goal is only partly known: professional judgment, leadership, empathy, and systems thinking require prolonged observation, multiple contexts, mentoring, and cumulative evidence [4].

The exploration circuit begins where the target state is not known in advance. It is triggered by curiosity, a hypothesis, an anomaly, a personal goal, an open-ended experiment, or an opportunity for which no employer yet exists. The graph must be capable not only of finding paths between known nodes, but also of creating new nodes and connections [5].

Circuit	Type of goal	Primary mechanism	Role of the digital system
Adaptation	The target action is known	Simulation, action, criterion, authorization	High
Development	The direction is known; the final state is refined over time	Observation across multiple contexts, mentor, portfolio	Contextual
Exploration	The outcome is not known in advance	Hypothesis, experiment, creation of a new graph node	Supportive, not directive

THREE MODES OF THE EDUCATIONAL GRAPH: ADAPTATION, DEVELOPMENT, EXPLORATION

The graph supports different forms of human change; it does not reduce them to one mechanism.

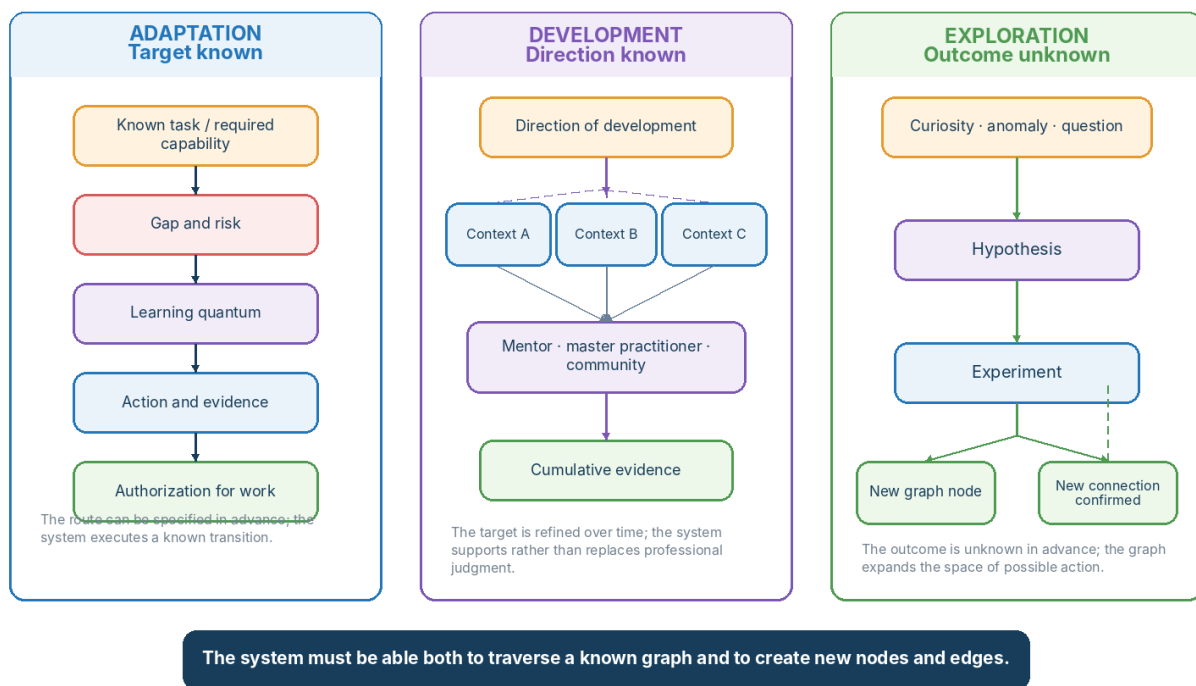


Figure 6. Three modes of the educational graph. Adaptation follows a known route, development progressively refines the target state, and exploration expands the space of possibilities.

The adaptation circuit prepares a person for the world as it exists. The exploration circuit enables a person to change the world itself.

Human actors regulate different parts of the same transition; none is the sole source of truth.

Actor	Role in the learning transition	What the model must represent
Student	Acts, selects a pathway, requests assistance, and contests decisions	Capabilities, goals, contexts, evidence, gaps, authorizations, and control over disclosure
Teacher	Designs the capability, quantum, simulation, and criteria; resolves complex cases	Expertise, scope of authority, methods, quanta created, results, and diagnostic accuracy
Mentor	Reduces dependence on assistance during the transition to independent action	Interventions, amount of assistance, change in independence, and boundaries of responsibility
Expert	Verifies ambiguous and critical results	Domain of competence, authority to verify, decision history, disagreements, and appeals
Employer	Defines work requirements, risk, and authorization; returns actual results	Tasks, processes, criteria, critical errors, simulation data, and work outcomes
Educational organization	Regulates the portfolio of transitions and the allocation of resources	Actors, capabilities, constraints, cost, speed, and durability of results
AI actor	Generates hypotheses and locally adapts the intervention	Context, permitted operations, confidence, and decision log; no unilateral power to assign an irreversible status

10. The Student as an Actor

In Simulator, each student is represented as a distinct actor with a unique actor_id. The student's digital twin persists across the educational and professional trajectory, but does not claim to describe the person's identity or personality. The model connects demonstrated capabilities, goals, roles, work contexts, current authorizations, unresolved gaps, and the history of semantically significant learning events. It records not only the result, but also the degree of independence and the assistance without which the result would not have been achieved.

The student is not a passive object of intervention. The student accepts or rejects a goal, selects an available pathway, requests assistance, submits evidence, contests an interpretation, and controls data disclosure. The digital twin must strengthen the student's agency, not replace it.

11. The Teacher as an Actor

The teacher must not exist in the system merely as a name on a schedule. The teacher's digital twin connects the domain of expertise, the authority to define and verify capabilities, the quanta created, the methods used, diagnostic decisions, and students' actual results. The teacher's function changes. Instead of repeating the same content for every cohort, the teacher designs capability models, simulations, and criteria; analyzes complex errors; mentors students; and develops the system of transitions itself. AI and Simulator assume routine adaptation and first-line feedback.

When a quantum designed by a teacher systematically fails to produce the stated transition, the system must

revise not only the student model, but also the models of the teacher, the method, the simulation, and the quantum itself.

12. The Mentor and the Expert

The mentor regulates the transition from assisted action to independent performance. The result is not a task completed jointly, but a reduction in the person's dependence on prompts and intervention. The system therefore records the form of assistance, the moment of intervention, and the development of independence. The expert enters the process where an automated rule cannot reliably assess professional judgment, an ambiguous result, or a critical capability. The expert verifies the decision within their domain of competence and assumes responsibility for the qualification claim.

Human assessment is not treated as infallible. The digital twins of mentors and experts include the boundaries of their authority, decision histories, disagreements, reviews, and the outcomes of appeals.

13. The Employer as an Active Participant in Learning

The employer is not an external customer who appears only after the student graduates. The employer is a full actor in the learning process. The employer provides the system with its most important information:

- which tasks actually exist
- which processes are changing
- which capabilities are required
- under what conditions they are applied
- which errors are tolerable
- which errors are critical
- what level of independence is required
- which tools are used
- what quality of result is sufficient
- when a previously demonstrated capability becomes obsolete.

13.1. The Employer Creates Requirements

A change in technology, process, product, legislation, or organizational role must automatically change the model of required capabilities. The employer should not send a university a list of competencies once every several years.

Its digital twin continuously publishes current requirements arising from actual work.

13.2. The Employer Supplies Tasks

Learning quanta should be based not only on artificial assignments, but also on real or anonymized work situations:

- documents
- operational data
- projects
- incidents
- errors
- decisions
- process changes
- new customer requirements.

13.3. The Employer Provides the Context for Simulation

The employer's digital twin makes it possible to reproduce:

- the process structure
- roles
- input data
- constraints
- tools
- rules
- possible errors
- the consequences of decisions
- criteria for the result.

This model is used to create a safe simulation of real activity.

13.4. The Employer Participates in Assessment

The learning system can verify an action in a simulation. But transfer of the capability to real work must be verified by an actual result. Assessment may include:

1. a learning exercise
2. a simulation
3. performance under observation
4. limited authorization
5. independent performance
6. employer confirmation of a sustained result.

13.5. The Employer Manages Authorization

A demonstrated capability must affect a person's real rights:

- to observe
- to perform with prompting
- to perform under supervision
- to perform a standard task independently
- to perform a higher-risk task
- to assess the actions of others
- to modify the process itself.

The employer determines the level of evidence required for each authorization.

13.6. The Employer Closes the Feedback Loop

When an employee successfully completes a learning quantum but cannot perform the real task, the defect may lie in:

- the capability model
- the simulation
- the assessment criterion
- the learning quantum
- the teacher's evaluation
- the employer's requirements.

The actual work result must return to Simulator and change the models of all connected actors. The employer is not a consumer of the educational result, but a co-author and active regulator of the learning process.

THE EMPLOYER IN THE COMPLETE LEARNING-TRANSITION CIRCUIT

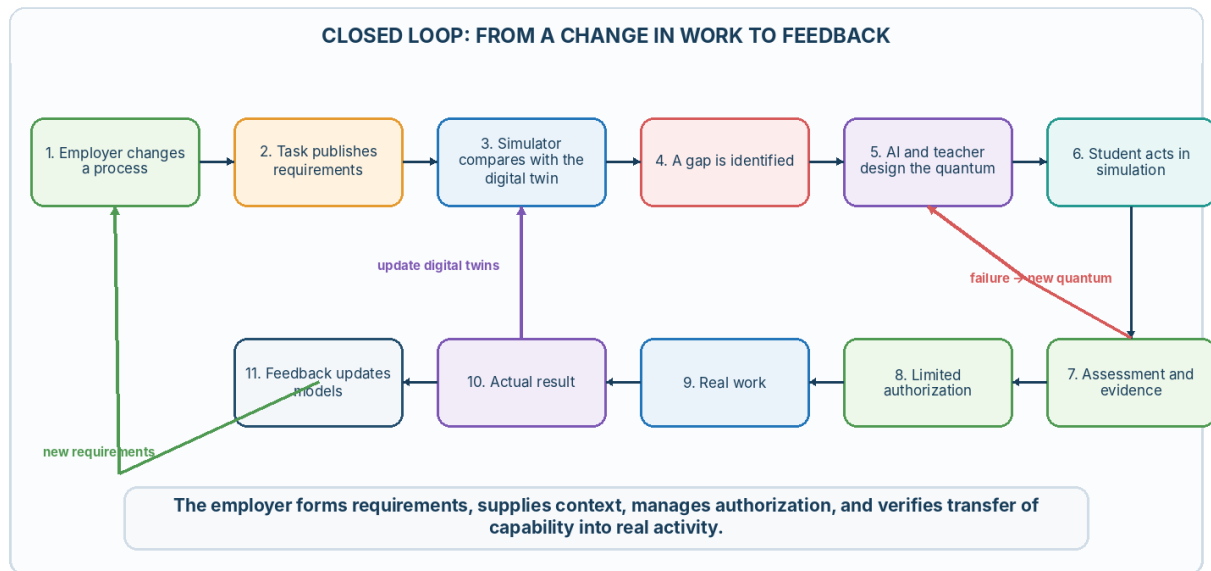


Figure 7. The employer in the complete learning-transition circuit. The employer forms requirements, supplies context, participates in assessment, manages authorization, and returns the actual result to the educational system.

13.7. The Employer Creates a Market for Transitions

Under structural unemployment, publishing a job title and waiting for a “ready-made” candidate is insufficient. The employer must publish a machine-readable model of the unfilled work: tasks, required capability states, conditions, risk, admissible assistance, quality criteria, and authorization rules. Simulator compares this model not with the title of the person’s previous occupation, but with the graph of their demonstrated capabilities. It identifies transferable nodes, missing transitions, and blocking constraints. One candidate may be closer to the role because of industry experience, another because of tool-related capabilities, while a third may have no safe or economically rational pathway to it.

The employer can reserve a position for a candidate, co-fund specific quanta, provide work data and a simulation, appoint a mentor, issue provisional authorization, and verify the transition through the actual result. Learning then becomes demand-led rather than the production of programs in the hope that the market may someday need them.

The employer’s task changes: it is no longer only to search for an ideal candidate, but also to participate in creating a demonstrable transition from available human capital to unfilled work.

13.8. The Minimum Employer Model and Levels of Maturity

A complete digital twin of the company should not be an entry requirement for the system. The first transition requires only one real task to be described: the result required, the inputs the person receives, the errors that are critical, the actions that are permitted, the evidence considered sufficient, and the actor who decides on authorization.

The rest of the model can be constructed gradually from work documents, instructions, incident logs, actual results, and mentor observations. The employer model is a testable hypothesis: when people who formally satisfy the requirements consistently fail in the work, the requirements, simulation, or criteria must be corrected—not only the person.

Level	What the employer provides	What becomes possible
0. Buyer of a job-ready candidate	A job title and the expectation that the market will supply a fully prepared person	Traditional recruitment only; no individual transition is computed
1. Basic	One task, the required result, input data, and critical errors	Computation of the first individual gap
2. Operational	Cases, assessment criteria, a mentor, and limited authorization	Simulation and verification of transfer to work
3. Process-based	A machine-readable role, related tasks, tools, and process versions	Automatic graph updates and reskilling at scale
4. Systemic	An event-driven twin of processes, risks, results, and resources	A continuous market for transitions within and between organizations

14. The Educational Organization as an Actor

A university or corporate learning system must have its own digital twin. This requirement follows from the Good Regulator Theorem and from the need to maintain a current model of the organization's own capabilities, constraints, and results. If an organization attempts to manage changes in capability without models of:

- students
- teachers
- employers
- tasks
- capabilities
- learning quanta
- assessments
- its own resources and constraints,

then it is not a good regulator. It may conduct classes, assign tasks, administer examinations, and issue diplomas, but it cannot demonstrate that it manages changes in capability. The educational organization's digital twin must represent:

- which transitions it can provide
- for which students
- at what speed
- at what cost
- with what probability of success
- with which teachers
- in which simulations
- under which criteria
- with what durability of the result
- where its constraints lie
- which quanta do not work
- where the system depends on individual experts
- which employer requirements remain unmet.

The central question changes.

It is no longer:

How many courses do we deliver?

It becomes:

Which changes in capability can we demonstrably produce?

15. Why an LMS Is Not a Digital Twin

A modern learning management system usually stores:

- | | |
|----------------------|-----------------|
| • a list of students | • attendance |
| • a course catalog | • grades |
| • materials | • test results |
| • a schedule | • certificates. |

This is an administrative model of educational activity. It shows what was assigned to a person and what the person formally completed. But it does not show:

- | | |
|--|--|
| • what the person can do | • the context in which the result was achieved |
| • how the person's capability changed | • how durable the result is |
| • which intervention caused the change | • what is required for the next transition |
| • what assistance the person used | • whether the learning corresponds to real activity. |

An LMS represents the document flow of the old educational system. A digital twin must model the causal circuit through which capabilities change.

Part III. Executable Entities and Simulator

Executable entities turn an educational description into a causal transition circuit.

Entity	What it records	Critical behavior
Task	Goal, inputs, required result, deadline, constraints, risk, tools, and criteria	Presents requirements and connects work to the person's state
Capability	Action, context, quality, independence, assistance, and period of validity	Has a current state and a required state
Gap	Difference between the current and required states, and the criterion for closure	Triggers a specific transition rather than a generic course
Quantum	Initial and target states, intervention, assignment, assistance, branches, and assessment	Completes only when sufficient evidence has been created
Attempt	Sequence of actions, tools, assistance, time, errors, and result	Distinguishes independent capability from a result produced on the person's behalf
Error	Expected and actual action, context, consequences, and hypotheses about causes	May reveal a defect in the assignment, simulation, criterion, or teacher
Evidence	Action, context, assistance, criteria, assessor, date, and scope of applicability	Supports a qualification claim and can be transferred between systems
Authorization	Rights to act, level of risk, constraints, and validity period	Can be expanded, restricted, suspended, or revoked
Simulation	A safe model of the real work context	Assesses action without exposing the production system to risk

16. The Task as an Actor

A task is more than a textual description. As an actor, it contains the goal, input data, required result, deadline, constraints, risk, admissible assistance, tools, assessment rules, and consequences of error.

The task connects the employer's digital twin with the student's digital twin. A person's readiness is always computed relative to a specific task, not an abstract occupational title.

17. The Capability as an Actor

Every consequential capability is represented by an operational model: action, object, context, tools, constraints, quality criteria, complexity, independence, admissible assistance, verification rules, and period of validity.

The capability connects task requirements, the student's state, learning interventions, attempts, assessments, evidence, and authorizations. A skill label without these connections is not a sufficient model.

18. The Gap as an Actor

A gap must not remain a textual recommendation generated by AI. It exists as an explicit difference between the current and target states of a specific capability, tied to a person, task, time, risk, and closure criterion.

Example of a gap

The student can construct a financial model from a prepared template, but the task requires the student to select the structure independently and test the robustness of the assumptions.

This is not a generic “gap in financial modeling.” It is the precise reason why the person cannot yet receive the required authorization.

19. The Learning Quantum as an Executable Actor

In Simulator, a learning quantum is not a fragment of content but an executable transition process. It connects the initial and target states, the intervention, the person’s action, admissible assistance, branches following an error, assessment rules, and the required evidence.

An explanation, example, exercise, simulation, prompt, or consultation may form part of a quantum, but none of these elements is a quantum on its own. The criterion of a quantum is a demonstrated change of state, not the small size of the material.

20. The Attempt as an Actor

A capability cannot be assessed solely from the final answer. A semantically significant attempt records the sequence of actions, tools used, interactions with AI, mentor assistance, intermediate decisions, time, errors, corrections, and assessment result. The final answer may be identical while the capability state differs: the person may have acted independently, after one prompt, after a complete explanation, or with the aid of an AI system that effectively solved the task on the person’s behalf.

At the same time, every click, pause, and draft must not become a permanent part of the digital twin. The level of detail and retention period are determined by the purpose, risk, and rules of the private learning circuit.

21. The Error as an Actor

An error must not disappear inside a negative grade. A consequential error becomes an event when it changes the learning pathway, risk level, or authorization decision; it records the expected and actual action, context, consequences, and hypotheses about causes.

An error may indicate not only the student’s state, but also a defect in the task, instruction, simulation, criterion, teacher, employer model, or educational process itself. Its function is therefore to trigger diagnosis and correction, not automatic punishment.

22. Evidence as an Actor

Every verified transition must have evidence: the result of real work, an action in a simulation, a document, a software artifact, a decision made, a series of consistently performed operations, an automated assessment, an expert evaluation, or a combination of these.

Evidence specifies what is being demonstrated, who acted, in what context, with what assistance, against which criteria, by whom it was assessed, when it was obtained, and where it applies. A certificate may be generated from evidence. A certificate does not replace evidence.

23. Authorization as an Actor

Authorization determines which actions a person has the right to perform in a real system. It is created on the basis of evidence and may be expanded, restricted, suspended, revoked, or made subject to reverification.

Authorization connects learning with activity and risk. The complete progression from observation to the right to modify the process is presented in Section 26.

24. AI as an Actor Within Simulator

AI plays an important role in quantum learning. It can:

- diagnose a probable cause of error
- select an explanation
- generate an example
- create an exercise
- adjust difficulty
- simulate a dialogue
- analyze an attempt
- provide feedback
- propose the next quantum
- recommend involving a teacher or expert.

But AI is not the source of truth about the student's state. A large language model is probabilistic. A digital twin must be precise, reproducible, and verifiable.

AI may formulate a hypothesis:

The probable cause of the error is a misunderstanding of the liquidity constraint.

But the capability state does not change because AI has declared it so. It changes only after the prescribed assessment procedure and the creation of evidence. AI generates hypotheses and interventions. Simulator applies rules, executes processes, and records transitions.

The digital twin stores state.

24.1. Data, Evidence, and the Boundaries of Observation

The digital twin must separate temporary technical telemetry, the private learning history, portable evidence of capability, and the data required for a specific authorization. A draft, an unsuccessful hypothesis, or a request for assistance must not automatically become a permanent element of an employment record.

The private learning circuit stores attempts, difficulties, prompts, and reflection only to the extent and for the period required by the current transition. The portable circuit contains the demonstrated action, context, quality, independence, date, period of validity, and source of verification.

Boundary	Mandatory rule
Bounded model	The twin describes only states relevant to a specific purpose; it is not a personality profile.
Control and portability	The person can view the model, obtain a portable copy of the evidence, and control disclosure.
Data minimization	The employer receives the necessary evidence and authorizations, not the complete learning log.
Protected learning	A learning error does not automatically become an employment, reputational, or social sanction.
Retention period	Raw attempts and errors are retained for a limited period and do not form a permanent rating.
Correction and appeal	The person can add context, correct data, and request human review.
Decision accountability	AI does not unilaterally assign an irreversible status or revoke authorization without due process.

LEARNING HISTORY AND PORTABLE EVIDENCE

The employer receives sufficient confirmation of capability—not the person's complete learning history.

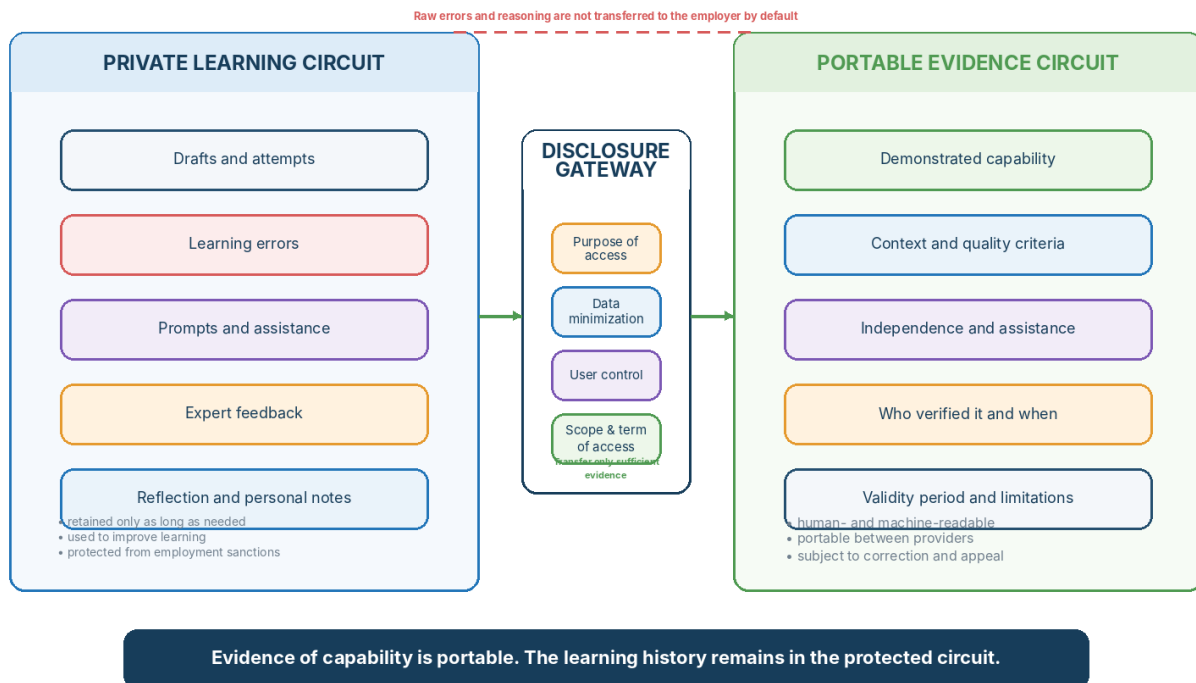


Figure 8. Separation of the private learning history from the portable evidence circuit. The employer receives sufficient evidence of capability, not a complete log of errors, prompts, and drafts.

A learning error must remain part of a safe space for development and must not automatically be used as grounds for restricting future opportunities.

25. Simulator as an Execution Environment for Learning

Simulator is not needed to visualize a learning trajectory. Its function is to execute it. Simulator is the Corezoid Inc. platform on which the author is implementing this architecture. The concept itself is technologically neutral and may be implemented in any event-driven environment that supports actor identity, state, rules, transactions, evidence, and auditability. Simulator simultaneously contains the digital twins of:

- | | | |
|-----------------------------|-------------------|------------------|
| • Students | • Work processes | • Errors |
| • Teachers | • Tasks | • Evidence |
| • Mentors | • Capabilities | • Simulations |
| • Experts | • Gaps | • Rules |
| • Employers | • Learning quanta | • Authorizations |
| • Educational organizations | • Attempts | • AI actors |

Every change occurs through an event and a transaction. A complete cycle may proceed as follows:

- | | |
|---|---|
| 1. The employer changes a work process. | 11. The attempt is recorded as a separate actor. |
| 2. The employer's digital twin publishes new requirements. | 12. The result is assessed. |
| 3. A task actor is created or changed. | 13. Evidence is created. |
| 4. The task presents requirements for capabilities. | 14. The capability state changes. |
| 5. Simulator compares them with the student's digital twin. | 15. The employer issues limited authorization. |
| 6. A gap is identified. | 16. The actual result returns to the system. |
| 7. A gap actor is created. | 17. The capability is confirmed, restricted, or a new gap is created. |
| 8. AI proposes a learning quantum. | 18. The models of all connected actors are updated. |
| 9. A teacher, rule, or expert confirms its applicability. | |
| 10. The student performs the action in a simulation. | |

This is no longer a content delivery system.

It is an executable circuit for managing human development.

25.1. An End-to-End Example: Marina's Transition to a New Role

Marina is 33. She works as a contact-center agent for a service company. AI already handles standard questions, completes templates, and reduces the volume of routine inquiries. At the same time, the company has been unable for several months to fill a field-service dispatcher role.

The system does not declare Marina "redundant" or assign her a new occupation. It first records her voluntarily chosen goal, constraints, and transferable capabilities. Marina has already demonstrated communication under pressure, effective use of a CRM system, classification of requests, adherence to shift protocols, and correct

escalation of conflicts. The new role requires capabilities she does not yet have: technical terminology, incident prioritization, safety rules, and the dispatch protocol.

Transition	Scope	Sufficient evidence
Quantum 1	Technical terminology—6 hours	Correctly classifies 20 standard fault descriptions
Quantum 2	Safety rules—8 hours	Completes five simulations without a critical error
Quantum 3	Incident prioritization—10 hours	Completes three timed simulations at the specified quality level
Quantum 4	Dispatch protocol—8 hours + 16 hours with a mentor	Completes two supervised shifts and one independent shift

In the second simulation, Marina incorrectly classifies a potentially dangerous incident. The error does not disappear into a “fail” mark: it triggers causal analysis and a corrective quantum. The employer does not receive a permanent log of her errors—only current evidence of readiness and the corresponding level of authorization.

Indicator	Illustrative value	Assumption
Cost of the unfilled role	UAH 180,000 per month	Lost productivity, overloaded shifts, and delayed field visits
Cost of the individual pathway	UAH 60,000	Simulations, expert, mentor, and paid learning hours
Time to limited authorization	3 weeks / 48 hours	Learning is combined with current work
Estimated payback period	Approximately 10 working days	Assuming that the unfilled role genuinely costs the company UAH 180,000 per month

After receiving limited authorization, Marina moves into the field-service dispatcher role. In this illustrative scenario, her monthly income rises from UAH 28,000 to UAH 36,000, and the next transitions open in her individual graph: analyzing recurrent causes of incidents, designing automation scenarios, and managing the quality of the service process.

This calculation is illustrative and is not based on market statistics. Its purpose is to show that a quantum is evaluated not only pedagogically, but also economically: the cost of the pathway is compared with the cost of the unfilled role, the risk of error, and the time spent waiting for a “ready-made” candidate.

26. Simulation Before Authorization for Real Work

Not every transition can be assessed safely within a production process. An error by a physician, pilot, financial operator, engineer, or infrastructure administrator may be extremely costly. Simulator must therefore create a safe, executable model of the work context. The student receives not an abstract test, but a simulation of a real task:

- with the same input data
- with the same constraints
- with the same roles
- with the same possible errors
- with the same criteria for the result
- but without risk to the real system.

After successful performance, the person may progress through levels of authorization:

- | | |
|--|--|
| 1. observation | 5. performance under changed conditions |
| 2. performance with prompting | 6. performance of a higher-risk task |
| 3. performance under supervision | 7. the right to assess others |
| 4. independent performance of a standard operation | 8. the right to modify the process itself. |

Part IV. The Individual Learning Graph

27. The Individual Learning Trajectory Is a Graph

A student's learning trajectory is not a linear sequence of courses, modules, and semesters. It is an individual graph of states and transitions. Its nodes are:

- | | | |
|---------------------|-------------------|------------------|
| • capability states | • learning quanta | • authorizations |
| • goals | • attempts | • work outcomes. |
| • tasks | • errors | |
| • gaps | • evidence | |

Its edges are transitions:

- | | | |
|-------------------------------|------------------------------------|--------------------------------|
| • acquisition of a capability | • correction of an error | • restriction of authorization |
| • increased independence | • restoration of a lost capability | • change of goal |
| • transfer to a new context | • receipt of authorization | • transition to a new task. |

A traditional trajectory looks like this:

Course 1 → Course 2 → Course 3 → Examination.

A quantum trajectory looks different:

Current state → identified gap → possible quantum → attempt → result → new state → new branch.

28. Every Student's Graph Is Unique

Even when two students are preparing for the same role, their trajectories should not coincide automatically. They differ in:

- | | | |
|------------------------|--------------------------------|-------------------------|
| • initial capabilities | • rate of learning | • available tools |
| • prior experience | • level of assistance required | • employer requirements |
| • demonstrated actions | • professional goals | • admissible risk. |
| • errors | • work context | |

One student may need only one simulation. Another may require an explanation, a series of intermediate tasks, and work with a mentor. A third may already possess the required capability and should proceed directly to assessment.

Assigning the same program to everyone means ignoring their actual states.

29. The Map Is Shared; the Route Is Not

An educational organization may maintain a shared map of the capability space:

- which states exist
- which transitions are possible
- which dependencies have been established
- which quanta are available
- which evidence is required
- which requirements are imposed by roles and tasks.

But the specific route is constructed individually. The common program is replaced by a common graph of possible transitions. An individual subgraph is constructed for each student. The shared map describes the available capability states, dependencies, and admissible transitions between them.

The individual graph is the current subgraph of that map generated for a particular person at a particular moment in time. The individual graph depends on:

- the student's current state
- the student's goals
- the requirements of the activity
- available actors
- evidence
- the moment in time.

It is not selected once. It is recomputed after every event.

30. The Graph Branches, Loops Back, and Develops in Parallel

A single gap may be closed through different pathways. The student may:

- receive an explanation from AI
- study an example
- complete a simulation
- perform the task with a mentor
- attempt the real task immediately under supervision
- submit existing evidence of the capability.

The system selects the pathway with regard to:

- the size of the gap
- risk
- cost
- available time
- previous results
- employer requirements
- the expected durability of the transition.

The graph permits reversals. A capability may:

SAME GOAL—DIFFERENT INDIVIDUAL GRAPHS

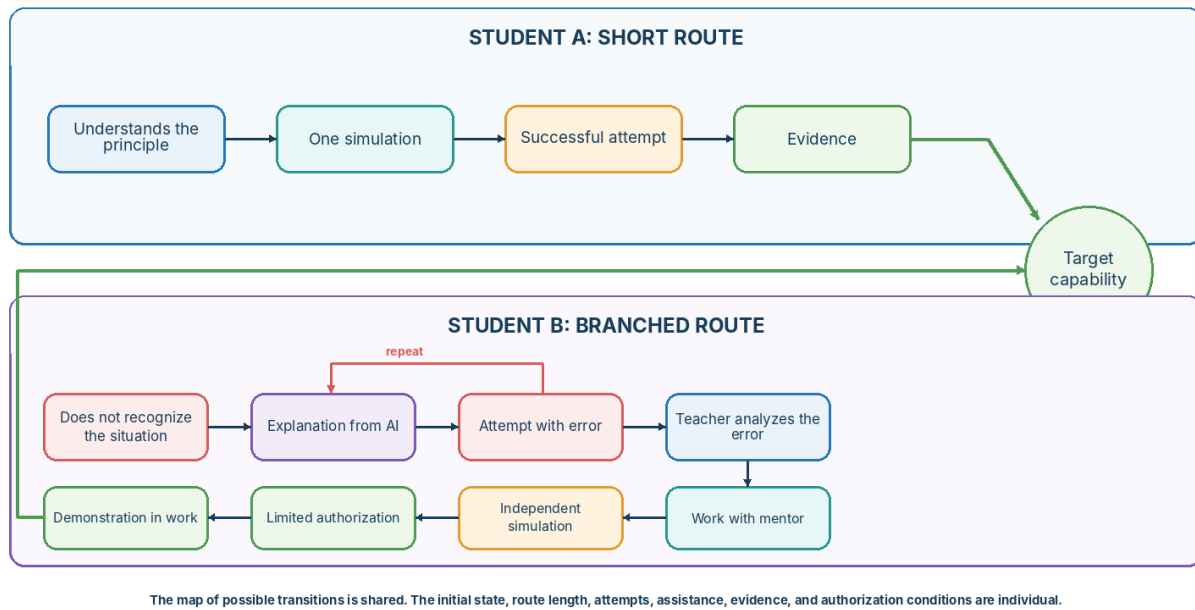


Figure 9. Different individual graphs for the same target state. The map of possible transitions is shared; the route is determined by the initial state, errors, assistance, risk, and evidence.

- weaken
- become obsolete
- fail to transfer to a new context
- prove to have been verified incorrectly
- lose relevance after a tool changes.

The following transitions may then occur:

Verified → *reverification required*. *Independent performance* → *temporary supervision*. *Full authorization* → *limited authorization*.

The graph also contains parallel branches. A person may simultaneously develop:

- proficiency with a professional tool
- decision-making capability
- communication
- risk management
- industry-specific knowledge
- capability to work in a new role.

An educational trajectory is therefore not a chain, but a multilayer graph of parallel transitions.

31. The Employer Influences the Individual Graph

The employer does more than define the final qualification. The employer changes the person's graph in real time. A new task creates a new goal node. A process change makes a previously demonstrated capability obsolete.

A critical error creates a mandatory reskilling branch. A successful work result may close a quantum without a separate examination. A new appointment opens a pathway to another role. The individual trajectory is therefore formed jointly by:

- the student's current state
- goals
- the map of educational opportunities
- teachers' actions
- mentor support
- AI decisions
- employer requirements
- actual results of activity.

A cross-occupational transition is a special case. A role for which demand is declining and a shortage occupation may have graphs that overlap only partially. Simulator must identify transferable capabilities, construct the missing branches, and account for the constraints of the particular person. The system does not impose an occupation. It computes a set of realistic pathways and presents the cost, duration, and risk of each honestly.

The individuality of a trajectory is therefore determined not only by different rates of learning, but by differences in achievable professional states. For one person, the shortest transition may be to a new office role that uses prior experience; for another, to a technical occupation following simulation and authorization; for a third, to entrepreneurial or project-based activity.

32. The Student's Digital Twin and Individual Trajectory

The student's digital twin and individual trajectory are not the same thing. The digital twin represents the person's entire current and historical state within the system. The individual trajectory is part of the digital twin. It contains:

- the pathway already completed
- active quanta
- unresolved gaps
- available next transitions
- blocked branches
- mandatory assessments
- possible goals
- projected pathways
- connections with employers and tasks.

The trajectory is not stored as a list of assigned courses.

It is executed as a living graph of actors.

33. How Quantum Learning Differs from Personalized Learning

Contemporary systems call the following "personalization":

- selecting a pace
- recommending the next video
- selecting material
- changing test difficulty
- adapting the interface.

This personalizes the consumption of content. Quantum learning personalizes the graph of transitions itself. The following may differ:

- the initial node
- the target state
- the sequence of quanta
- the number of attempts
- the level of assistance
- assessment rules
- evidence requirements
- the teacher's participation
- the employer's participation
- authorization conditions
- the connection to real work.

What is individualized is not the way a common program is completed.

The common program disappears as a mandatory route.

34. The Educational System as a Graph of Graphs

The student's individual graph does not exist in isolation. It is connected to the employer's task graph, the capability graph, the graph of learning transitions, the graph of teachers' expertise, the evidence graph, the financing graph, and the graph of access rights.

Each of these graphs has its own nodes, owners, rules, and update frequency. The educational system is therefore a graph of interconnected graphs, not a single centralized map.

Graph	Primary nodes and connections	Effect on the individual pathway
Person	Capabilities, goals, constraints, evidence, and authorizations	Determines the initial state and achievable goals
Work	Tasks, roles, tools, risk, criteria, and outcomes	Defines required states and authorization rules
Transitions	Quanta, simulations, mentors, experts, and alternative branches	Offers possible pathways between states
Providers	Teachers, laboratories, capacity, cost, and availability	Constrains the time, price, and executability of the pathway
Evidence and accountability	Attempts, assessments, authors of claims, validity periods, and appeals	Determines confidence in the state and identifies who is accountable when a claim is contested
Financing	Payers, contracts, spending limits, and payment conditions	Determines the economic accessibility of the pathway
Rights and access	Consent, disclosure, portability, retention periods, and powers	Determines who can view data and make decisions

An event in one graph restructures the others. A new tool changes the work graph; this creates new target states, opens or closes quanta, changes the pathway's cost, modifies evidence requirements, and affects current authorizations. Conversely, a successful work result updates the person's graph and the quality models of the provider.

An educational pathway is simultaneously a graph of learning, accountability, payments, and access rights. Simulator enables the joint execution of these connections, but does not become the owner of the person or the sole source of truth.

Part V. Economics, Metrics, and Institutional Consequences

35. Metrics of Quantum Learning

Traditional indicators—hours, attendance, completed courses, tests, and certificates—describe the activity of the educational system, but do not demonstrate a change in capability.

Quantum learning requires not one long list of indicators, but five families of metrics associated with different decisions.

Metric family	Key indicators	Management question
Transition speed	Time from event to gap; from gap to evidence; from evidence to authorization; from a technological change to an available quantum	How quickly does the system convert a new requirement into readiness to act?
Reliability of evidence	Durability and transfer of capability; false verifications; correspondence between simulation and work; accuracy of teachers and mentors; proportion of successful appeals	Can the stated state and its scope of applicability be trusted?
Economics	Cost of the pathway and quantum; savings from recognized capabilities; ratio of pathway price to the cost of an unfilled role; scalability of teacher expertise; recurring B2B revenue	Who pays, what do they pay for, and where does the economic benefit arise?
Labor market and development	Time to new employment; cross-occupational transition rate; vacancies with executable pathways; proportion of transferred capabilities; new exploratory branches and nodes	Does the system reduce the structural gap and expand the space of possibilities?
Rights and transparency	Proportion of the learning history disclosed to the employer; portability of evidence; corrected decisions; retention periods; dependence on a single platform	Is efficiency being achieved at the cost of opaque power over the person?

The primary metric becomes not the completion rate, but the transition rate—the proportion of initiated transitions that end in a demonstrated, durable, and applicable capability within the specified context and time. This indicator must always be interpreted together with transition complexity, risk, and respect for the participant's rights.

36. The Economics of the Legacy Educational Unit

The course, semester, and degree persist not only because of pedagogical inertia. They support a stable financial, administrative, and regulatory system. The course has real advantages: it coordinates large groups, creates a shared intellectual environment, supports socialization, and makes it possible to construct a foundational sequence. Quantum learning does not abolish the course as a possible pedagogical container. It rejects the course as the primary unit of outcome, payment, and accountability.

The legacy system is rational relative to its own objectives—predictable funding, standardization, scheduling, reporting, and risk transfer. But these objectives are not identical to a demonstrated change in capability.

Actor	What the course provides as a unit	What the quantum model changes
University and its finances	Prepayment for a large package and predictable cash flow	Revenue from individual pathways, recurring transitions, and long-term stewardship
Regulator and accreditation system	Hours, plans, semesters, and credit units that are easy to inspect	Audit of capabilities, evidence, false verifications, and durability of results
Administration	Stable schedules, cohorts, staffing allocations, and classroom utilization	Dynamic allocation of teachers, simulations, and assessors across transitions
Teacher and department	A legible workload measured in contact hours and recurring cohorts	Compensation for intellectual assets, results, mentoring, and complex cases
LMS providers, publishers, and platforms	A catalogable product, licenses, and completion rates	Executable models, simulations, evidence, and open portability interfaces
Employer	Transfer of the initial training risk to the student and the state	Co-funding of role-specific transitions and responsibility for the work model
Student	A simple social signal, a cohort, a shared environment, and access to funding	Payment only for the necessary pathway while preserving the environment, foundations, and collective experience

The course persists not because it forms capability more effectively, but because cash flows, accreditation cycles, staffing models, and software products have been built around it. The transition to a quantum model will therefore be an institutional and economic conflict, not merely a methodological reform.

For the new unit to function, contracts, funding, accreditation, the model of teacher workload, and the distribution of risk among the student, university, employer, and state must all change.

36.1. The Diploma as a Social Signal and the Speed of the Market

Historically, the diploma has performed not only an educational function, but also a signaling function. Prolonged completion of a selective program told the market that a person could comply with complex rules, sustain a demanding workload, belong to a particular environment, and pass through a social filter. This function is real; quantum learning does not deny it.

But the economic value of a delayed signal depends on the speed of the world. While professions and technologies lasted for decades, an employer could use a diploma as a convenient but lazy proxy for general readiness. When work itself changes faster than the educational cycle, waiting five years becomes an excessively expensive way for the mass market to determine whether a person can perform a specific task today.

“The traditional university may continue to treat the diploma as a social filter. But the market will no longer wait five years for you to mature. Except in professions where the cost of error requires a long pathway, the employer needs evidence of capability here and now.” — Alexander Vityaz

The market therefore increasingly separates two distinct forms of value. Community, networks, culture, initiation, and affiliation with a strong university may retain independent value. But they must no longer substitute for a qualification claim. Belonging to a group is one product; demonstrated capability is another.

This shift is also reflected in anticipated hiring practices. According to the World Economic Forum, over the 2025–2030 horizon, 81% of employers expect to use work experience as a mechanism for assessing candidates, 48% expect to use direct skills assessments, and 43% expect to require a university degree [8]. The diploma does not disappear, but it ceases to be the only—or a sufficient—proxy for readiness to work.

Function	Logic of the diploma	Logic of quantum learning
Selection	Four or five years as an indirect signal of discipline, endurance, and compliance with rules	Direct evidence of capability to perform a specific class of tasks
Speed	The signal appears only after the entire program is completed	Evidence appears after each completed transition
Community and network	Membership in a group and brand is combined with the qualification claim	Community remains a distinct value but does not substitute for evidence of capability
Long-cycle professions	Duration and credit hours serve as the principal proxy for readiness	The long pathway remains, but consists of a sequence of evidence, simulations, and authorizations
Response to the market	The employer reassesses the graduate after the diploma has been issued	The employer receives current evidence in the stated context and can see its limitations

For the working adult, this shift is particularly clear. The person already has a professional history, responsibilities, a family, and limited time. Requiring another multiyear social ritual merely to obtain a new signal means ignoring both the person’s initial state and the speed of market change. The person needs a portable result from the next transition, not another initiation. The exception is not long-form university education as such, but transitions that objectively require prolonged maturation and repeated assessment: medicine, aviation, safety-critical engineering, fundamental science, and complex professional judgment. The quantum model need not shorten such pathways artificially. It makes them executable and traceable: thousands of connected quanta, simulations, observations, and graduated authorizations in place of one undifferentiated statement—“studied for five years.”

The social signal is therefore neither prohibited nor administratively devalued. It simply ceases to count as sufficient evidence of professional readiness where the market already requires direct, current, and contextual confirmation of capability.

37. The Economics of the Individual Educational Pathway

Changing the primary unit of learning inevitably changes the object of the transaction. A traditional organization sells a preassembled package: a program, semester, year of study, or degree. The student pays for the entire route regardless of what they can already do, which disciplines they will actually need, or whether a capability transition ultimately occurs.

In the quantum model, the map of educational opportunities remains shared, but the specific pathway is

constructed individually. The invoice must therefore be generated from the individual pathway: recognized capabilities are excluded, unselected branches are not charged, and the price includes only the quanta, simulations, mentoring, and assessment actually required.

The student pays not for the entire map of what the university knows how to teach, but for the pathway actually selected and demonstrably completed.

37.1. A Recognized Capability Reduces the Price of the Pathway

Prior experience, self-directed learning, and the results of real work must have direct economic value. If a capability has already been demonstrated through sufficient and current evidence, the corresponding graph node is skipped. The person neither repeats the learning nor pays for it a second time. This creates genuine recognition of prior learning: not the administrative transfer of a course credit, but the removal of an already completed transition from the time, risk, and price of the new pathway. Portability of evidence between universities and employers becomes a condition of competition, not an optional platform feature.

THE STUDENT PAYS FOR THE PATHWAY ACTUALLY COMPLETED, NOT THE ENTIRE MAP

Capabilities already demonstrated and branches not selected are excluded from the invoice.

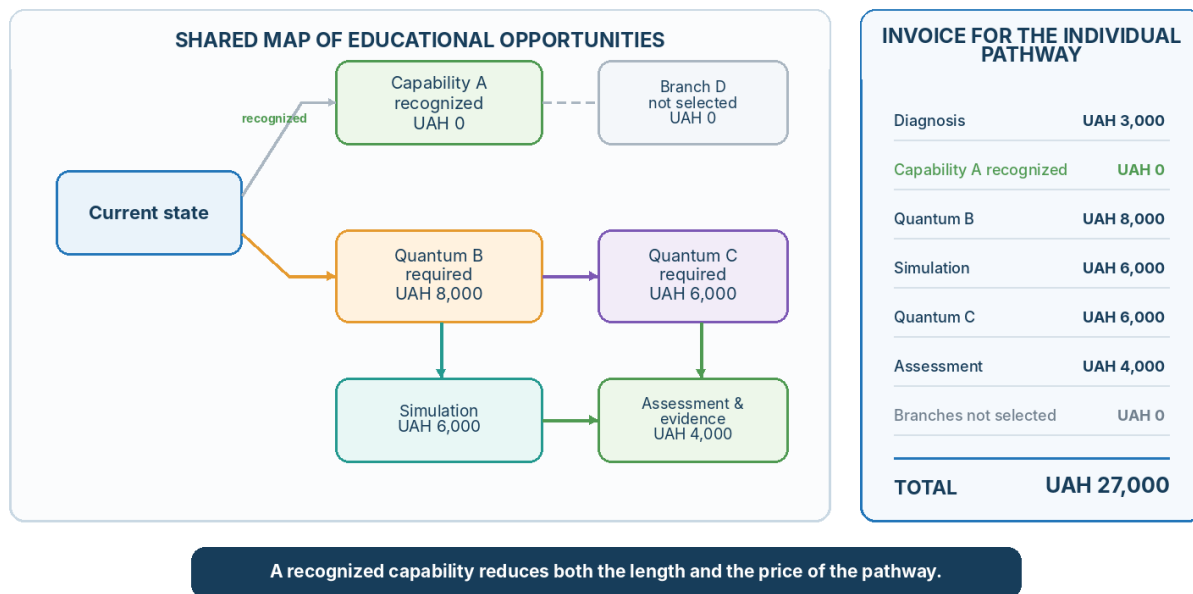


Figure 10. Economics of the individual pathway. The student pays for diagnosis, required quanta, resources used, and evidence; capabilities already demonstrated and branches not selected are excluded from the invoice.

37.2. What Exactly Is Paid For?

Payment for the actual pathway does not mean a microtransaction for every click, question, or error. Such a model would create an incentive to prolong learning and punish the person for needing additional support. The economic unit remains a substantively complete transition, and a reasonable number of attempts and instances of feedback is included in its price.

Pathway element	Pricing rule	Economic meaning
Diagnosis of the initial state	Paid once for the selected goal	Determines the initial node and excludes unnecessary learning
Capability already demonstrated	Zero in the variable part of the pathway	Reduces time and price; evidence must be portable
Learning quantum	Fixed price or payment after the transition is verified	The provider is responsible for the stated class of transitions, not for hours of content
Simulation or laboratory	Charged according to actual resource use	Separately accounts for expensive infrastructure and risk
Mentor or expert	Charged according to use, subject to a pre-established limit	Directs human attention to where it is genuinely necessary
Independent assessment	Separate price for the evidence	Separates the interests of the seller of learning from those of the assessor of the result

Part of the payment may cover unavoidable resources—a laboratory, simulation, or mentor’s work. The principal part should be tied to a verified result: the initial state has been recorded, the quantum executed, the new state demonstrated, and portable evidence created.

If the provider diagnoses the gap incorrectly or assigns a quantum that does not work, the student should not pay indefinitely for the provider’s own error. The contract must define a price ceiling, rules for recalculating the pathway, reassessment, and the allocation of risk.

37.3. Cost, Price, and Payer Are Different Concepts

The individual cost of a pathway does not mean that the student must pay the entire amount. The transition creates value for several parties: the person obtains a portable capability, the employer meets an operational need, the state reduces structural unemployment, and the industry alleviates a labor shortage.

Payer	What it is economically rational to fund	Possible form
Student	Portable capabilities that increase long-term freedom of choice	Personal payment, installment plan, learning account
Employer	Role-specific quanta, simulations, a mentor, and authorization	Corporate contract or co-funding of the pathway
State / employment fund	Transitions that reduce structural unemployment and social costs	Voucher or payment for sustained employment
Industry / association	Shared standards, shortage capabilities, and shared simulations	Industry fund or membership contribution
University / provider	A share of the risk associated with the quality of diagnosis and the quantum	Deferred payment after the result is verified

A single pathway may have several payers. A transition contract records each party’s share, payment conditions, the limit on the student’s personal expenditure, evidence criteria, rules for changing the pathway, and responsibility for defects in diagnosis, the quantum, or assessment.

To ensure that people with larger initial gaps do not pay more merely because they need more support, the system requires risk-adjusted funding, social vouchers, corporate co-funding, and a cap on out-of-pocket expenditure. The price of the pathway should reflect resources, but must not become a penalty for a weaker starting position.

37.4. The Quantum as an Outcome Contract and a Chain of Accountability

Today, a diploma is portable as a document, but responsibility for the qualification claim is almost entirely non-portable. The student pays for a program, the university issues a diploma, and the employer reassesses everything. When the graduate cannot perform the work, the claim dissolves among the program, teacher, examination, and the employer's hidden requirements.

In the quantum system, every capability claim has a provenance: task, context, attempt, assistance used, result, criterion, assessor, validity period, and scope of applicability. This makes it possible to identify not an abstract guilty party, but the element of the regulatory circuit whose stated condition or obligation was not fulfilled.

The provenance and responsibility model developed here extends the actor-accounting logic of *Management Debt*, which links observable events, decisions, omissions, traces, materialized risks, and consequences to accountable actors [11].

Failure	Accountable actor	Corrective action
Work requirements were described incorrectly	Employer or owner of the activity model	Correct the task model and recompute pathways
The initial state was determined incorrectly	Diagnostic provider	Repeat diagnosis at the provider's expense
The assigned quantum does not produce the stated transition	Learning provider and author of the quantum	Redesign the quantum, provide a compensatory pathway, reduce the reliability rating
The capability was verified incorrectly	Independent assessor	Review the evidence and the authority to verify
The simulation does not correspond to real work	Simulation owner together with the employer	Update the scenario and reassess
Events or evidence were distorted by the platform	Platform operator	Restore data, conduct an audit, and assume responsibility for integrity
The participant concealed material assistance or substituted the result	The participant	Invalidate the evidence through a transparent procedure
The work changed after verification	A new gap, not automatic fault on the part of the previous provider	Create a new requirement and quantum, and set a new validity period for the evidence

Accountability does not mean an absolute guarantee of employment success: a capability may weaken, the context may change, and a new task may fall outside the scope of the original evidence. But the system must show exactly what was promised, who assessed it, under what conditions, and why authorization was issued.

A course sells access to content. Quantum learning sells a verified transition and makes responsibility for the evidence traceable.

38. The Business Model of the Quantum University

The quantum university ceases to depend solely on one large sale at the beginning of a person's life. It becomes the operator of a continuous development graph: it diagnoses the state, recognizes what has already been demonstrated, supplies the necessary transitions, brings in experts and simulations, verifies the result, and returns when the technology, role, or goal changes again.

The one-time fee for a degree may decline because the student no longer pays for the same complete package as everyone else. But longer relationships, new categories of payer, and recurring revenue emerge: the person returns for new transitions, employers purchase continuous employee development, and the state purchases verifiable reskilling pathways.

38.1. The Working Adult Is the Primary Learner of the Quantum University

“The primary learner in the modern university is no longer the recent school graduate, but the working adult. This person does not come once for a diploma, but returns throughout an entire professional life—for the next verified transition in capability.” — Alexander Vityaz

The traditional university is designed primarily around a young person who can subordinate several years of life to the academic calendar. The quantum university is designed around an adult who already has an occupation, heterogeneous experience, and three recurring domains of responsibility: paid work, family—including care for loved ones and pets—and continuous learning. The person's time is limited, and the new capability must be applied in a specific context.

As Figure 1 shows, learning becomes one of several parallel circuits of responsibility for the working adult. The university's principal constraint is therefore not merely access to content, but its capacity to reduce the time, cost, and organizational friction of the transition without damaging the person's work, family, or recovery.

Parameter	Recent school graduate	Working adult
Initial state	A relatively common educational foundation	A unique graph of experience, demonstrated capabilities, and gaps
Time	Can follow the university calendar	Combines the third job—learning—with paid work, family, care for loved ones and pets, and recovery
Goal	A degree and entry into an occupation	The next role, technology, task, or professional transition
Context	Educational tasks and a common program	Immediate application in real work
Payer	Family, state, or student	Blended funding: person, employer, state, and industry
Product	Complete program	Individual pathway with recognition of what has already been demonstrated
Rhythm of the relationship	Several years followed by graduation	Repeated returns throughout the person's professional life
Criterion of success	Program completion and a diploma	Demonstrated capability, transfer to work, and new authorization

This change in audience produces a new operating model: year-round enrollment, short transition contracts, hybrid and asynchronous formats, recognition of prior experience, corporate subscriptions, shared funding, and billing based on the pathway actually completed. The university stops waiting for the next age cohort and begins supporting the economy already in operation.

38.2. From Selling a Degree to Stewarding a Graph

Parameter	Traditional university	Quantum university
Product	Degree or program	Individual graph of verified transitions
Unit of sale	Year, semester, course, credit	Diagnosis, quantum, simulation, mentoring, evidence
Relationship with the student	Several years, followed by graduation	Continuous stewardship throughout professional life
Role of the employer	Recipient of the graduate	Co-author of requirements, payer, and source of feedback
Teacher's work	Repeated lectures and manual assessment	Designing transitions, updating models, mentoring, and complex assessment
Revenue	Large payment for a package	B2C pathway, B2B subscription, payment for transition, simulation, and assessment
Quality	Academic performance and graduation	Speed, durability, and transfer of demonstrated capability
Primary asset	Content and brand	Capability maps, quanta, simulations, experts, and data on transition reliability

The university acquires a portfolio of interconnected products: core infrastructure and foundational environments, diagnosis, quanta, simulations, mentoring, independent assessment, stewardship of the individual graph, and corporate contracts. A student becomes not a participant in one program, but a long-term user of the university at different stages of professional life.

38.3. One Teacher Supports More Students

In the traditional model, a substantial share of the teacher's time is spent repeating the same lectures, explanations, and assessments for successive cohorts. Scaling requires either enlarging the audience or increasing contact hours. In the quantum model, the teacher's scarce expertise becomes a reusable intellectual asset. The teacher describes the capability, designs the quantum, creates the simulation scenario, and establishes the assessment criteria. AI and Simulator adapt the explanation, generate exercises, provide routine feedback, and route individual graphs.

A human remains in the circuit wherever automation cannot replace professional judgment: complex diagnosis, mentoring, research, conflicts between assessments, and verification of critical capabilities. One teacher can therefore support many more students without turning learning into a mass broadcast of identical content. The limiting resource becomes not the size of the audience, but the number of complex interventions that genuinely require human expertise. This allows the teacher to support more students while preserving quality where quality cannot be scaled automatically.

HOW THE TEACHER'S WORK SCALES IN THE QUANTUM MODEL

Scale comes not from removing the teacher, but from reusing expertise and managing exceptions.

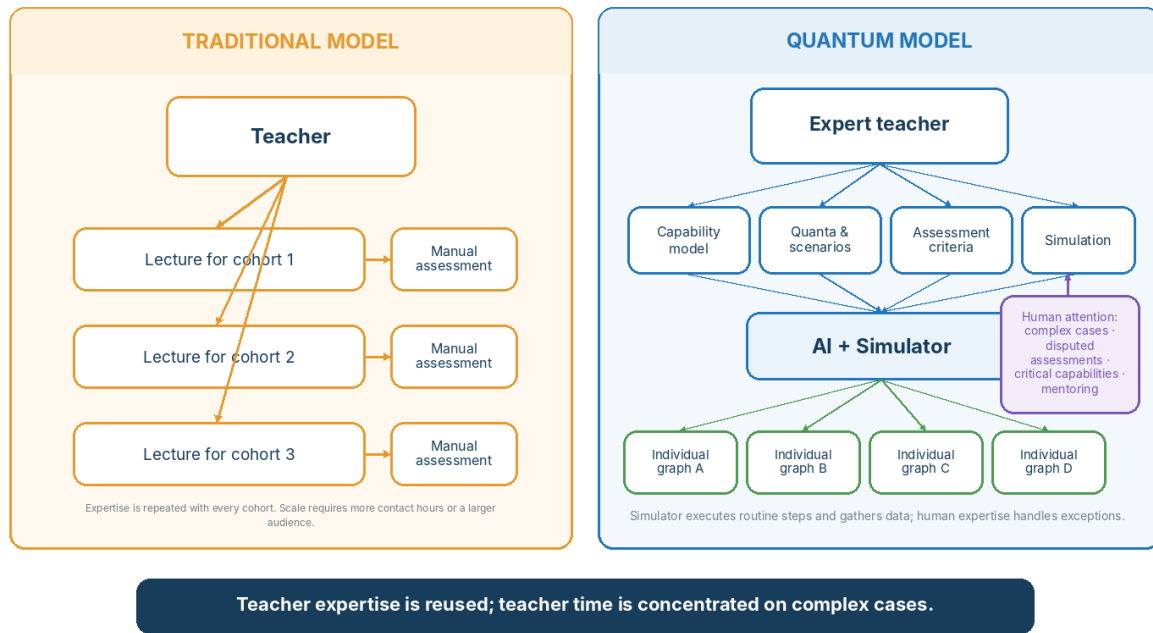


Figure 11. Scaling the teacher's work. In the traditional model, expertise is repeated for every cohort; in the quantum model, it is embodied in reusable models, quanta, criteria, and simulations, while human attention is directed to complex cases.

What changes	Description
Scaled through software	Standard explanations, examples, exercises, initial diagnosis, routine feedback, and routing
Reused as an intellectual asset	Capability model, quantum, simulation scenario, assessment criterion, and standard error analyses
Remains a scarce human resource	Mentoring, professional judgment, formulation of research questions, and complex, high-accountability assessment
New workload metric	Number of durable transitions and quality of escalations, not only contact hours and audience size
New compensation model	Payment for authorship, use of a quantum, a share of verified transitions, expert sessions, and mentoring

The teacher's income can likewise cease to depend exclusively on classroom hours. Possible forms include payment for authorship of a quantum, a licensing share for its use, compensation for durable transitions, expert sessions, and mentoring. The provenance of each quantum and its actual performance make the teacher's contribution measurable.

Scale must not be assessed only by the number of students per teacher. It must be normalized for transition complexity, durability of the result, the rate of false verifications, and the amount of human assistance. Otherwise, efficiency will once again be achieved at the expense of quality.

38.4. The University as a Continuous Learning Service for Companies

Companies already purchase university expertise after employees have formally graduated. In the practice of Corezoid Inc., we regularly commission university lectures in computer science, quantum computing, artificial intelligence, and related fields—to extend the preparation of new employees and reskill those who are already working. This demand shows that the relationship between the university and the company should not end when a graduate is hired. Today, however, such purchases usually remain a series of isolated events: one lecture, seminar, or short program. They rarely diagnose the individual gap, exclude what the person already knows, support application, or create evidence of the new capability.

The quantum university turns a one-time purchase of content into a continuous service. The company's digital twin publishes changes in technologies, processes, and roles; the system compares them with employees' graphs; the university offers only the necessary quanta, laboratories, simulations, and expert sessions; and the work result returns to the models of both organizations.

THE QUANTUM UNIVERSITY AS A CONTINUOUS LEARNING SERVICE FOR COMPANIES

A one-off lecture becomes a continuous circuit of gap diagnosis, learning, evidence, and capability renewal.

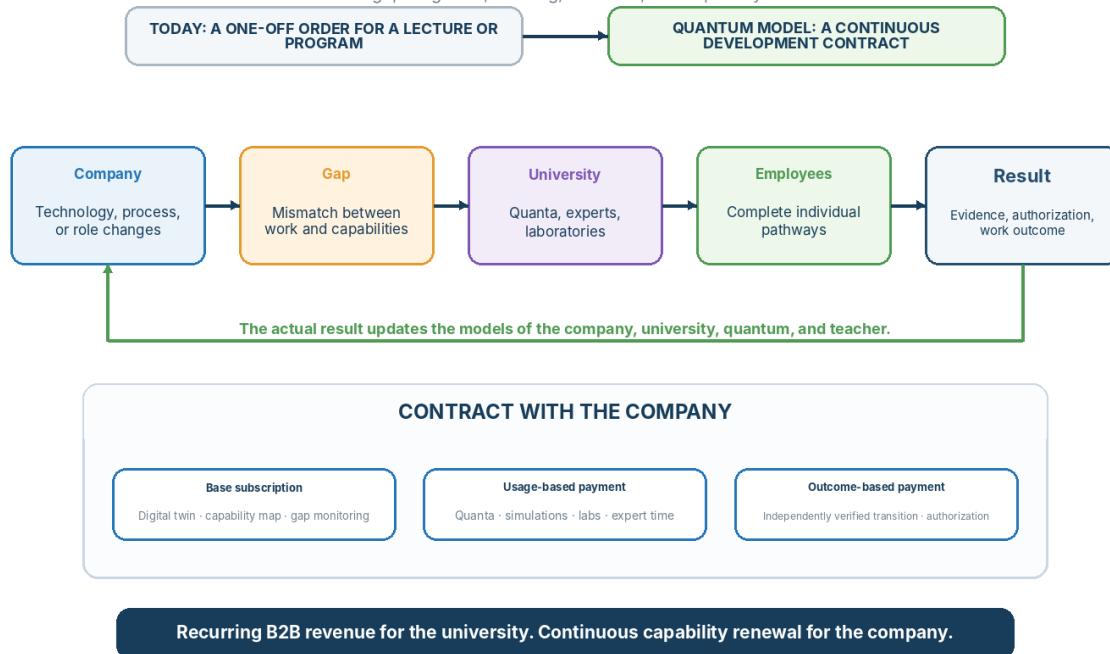


Figure 12. The university as a continuous learning service for a company. A change in technology or process creates a gap; the university forms individual transitions, and the actual work result updates the models of all participants.

A commercial contract may combine three layers: a base subscription for the digital twin, capability map, and gap monitoring; variable payment for quanta, simulations, laboratories, and expert time used; and an outcome-based component for an independently verified transition and authorization.

For the company, value is expressed not in the number of lectures delivered, but in shorter onboarding time, faster adoption of new technologies, lower cost of unfilled roles, fewer errors, and sustained currency of critical capabilities.

38.5. Revenue Sources and the New Cost Structure

Line of business	What the university sells
Diagnosis and pathway construction	Determination of the initial state, recognition of prior experience, and selection of an achievable goal
Quanta and transition libraries	Access to, or payment for, a transition actually completed
Simulations and digital laboratories	Use of expensive infrastructure without risk to the real system
Mentoring and expert assistance	Scarce human attention delivered through escalation
Independent verification	Creation of trusted and portable evidence
Continuous learning for companies	Base subscription, gap monitoring, graph updates, and variable payment for transitions
Government and industry contracts	Reskilling, filling shortage roles, shared standards, and simulations

The economics of the quantum university combines high initial costs for designing a high-quality capability model, quantum, and simulation with a lower marginal cost for each subsequent use. A common foundational quantum may serve employees of many companies, while a role-specific layer is added on top.

This model creates economies of scale, but also raises the importance of open standards and portability. If one platform holds the person's evidence and graph captive, the efficiency gain becomes monopoly rent. Competition is possible only when the student and company can change providers while retaining demonstrated states and the history of accountability.

39. The Role of the University

Quantum learning does not mean the destruction of the university. It means the end of the university's monopoly on preparing a person for professional life. The university may remain a space for:

- fundamental knowledge
- research
- laboratories
- scientific thought
- complex intellectual transitions
- independent assessment
- the formation of capabilities that cannot be acquired directly in the work process.

But the university must no longer be a diploma factory through which a person passes once at the beginning of life. It must become one of the actors in a continuous educational system. Its digital twin must show:

- which transitions it can provide
- for which students
- through which teachers
- under what conditions
- at what cost
- at what speed
- with what durability of the result
- on the basis of what evidence.

39.1. The University as an Operator of Long-Horizon and Exploratory Transitions

The university must support all three circuits. In adaptation, it forms and independently verifies professional transitions. In development, it creates long-horizon environments for mastery and for scientific and professional judgment. In exploration, it preserves an autonomous space for knowledge that does not yet have a vacancy, a customer, or a known return on investment.

Knowledge beyond immediate demand is not a defect. It creates optionality: the human capacity to formulate a question, build a new field, and change the graph of activity itself. The requirements of the current employer therefore cannot be the sole source of educational goals.

39.2. Transitions and Exploration Require Different Sources of Funding

Payment for an individual pathway is applicable where the target state and sufficient evidence can be substantively defined. It must not turn the entire university into a supplier of short-term functions for current business.

Fundamental knowledge, culture, research, and the creation of new fields are public and long-term infrastructure. They require core public funding, grants, endowments, research contracts, and internal cross-subsidization. Revenue from professional transitions and B2B services may strengthen this circuit, but must not replace it with the logic of immediate payback.

The quantum university is a hybrid: transitions are funded as a demonstrable service; exploration is funded as the creation of a future space of possibilities.

40. Why the Corporate University Does Not Solve the Problem

Corporate universities, postgraduate programs, online platforms, and course libraries usually preserve the previous architecture. They still offer:

- long programs
- preassembled modules
- a common pace
- learning separated from work
- verification of completion
- a certificate at the end.

This is not a new educational system. It is the old system inside a company. A digital platform does not make learning modern when the course remains the primary unit. Personalizing the interface is not personalizing the transition.

Reducing a lecture from one hour to fifteen minutes does not turn it into a learning quantum.

40.1. Digitizing Delivery Does Not Change the Educational Unit

“Coursera and similar services are lipstick on a pig. They digitized an obsolete process and continue to sell the same product: a preassembled course, a program, and a qualification document at the end—only online.” — Alexander Vityaz

Coursera and similar platforms have indeed solved an important delivery problem: they expanded access, reduced the marginal cost of distributing content, and made learning asynchronous and global. But Coursera’s official product portfolio remains organized around courses, Specializations, Professional Certificates, university certificates, and online degrees; its enterprise offering sells teams access to a catalog and analytics [7].

What has been digitized	What has not changed
Delivery: the classroom has been replaced by a browser, mobile application, and video	Primary unit: the course, specialization, certificate, or degree
Access: a global catalog, asynchronous participation, and subscription	Pathway: preassembled by the provider rather than computed from the state of a specific person
Control: tests, automated grading, and completion analytics	Result: completion and a credential, rather than demonstrated action in a work context
Scale: one body of content serves millions of users	Economics: access to a package or seat is sold, rather than the transition actually required
Enterprise product: a catalog and analytics for teams	Accountability: the connection among learning, evidence, and real authorization remains open

The mode of delivery has been digitized, but the object of the transaction has not. The user still purchases a preassembled educational package and receives a qualification document after completion. The platform knows what the person viewed and completed, but this alone does not demonstrate that the person can perform specific work in a specified context, at the required quality and level of independence. An online platform can become part of quantum learning, but only if it changes the unit: from a catalog of courses and credentials to gap diagnosis, an individual graph, simulation, evidence, and authorization. The criticism concerns neither the digital channel nor Coursera as a company, but the architecture of education that the digital channel has preserved.

A university’s continuous B2B service is not simply a new name for a corporate university. Its unit is the individual verified transition rather than the company’s internal program. The university preserves the independence of assessment, works with multiple employers, reuses a common foundation, and returns actual results to its own quality model.

Part VI. Principles, Definition, and Manifesto

41. Principles of Quantum Learning

Capability matters more than knowledge. Knowledge has value when it enables a person to act, understand, create, or make a decision. **The transition matters more than the content.** Content is a means; the result is a change in the person’s state. **The event matters more than the schedule.** Learning begins when a task, goal, error, hypothesis, or new context emerges. **Action and evidence matter more than completion and certification.** A claimed capability must be supported by a relevant result.

The individual graph matters more than the common program. Initial state, route, assistance, and evidence differ from one person to another. **The model and feedback matter more than administrative representation.** The digital twin is updated by events and corrected by actual results. **A recognized state matters more than repeated learning.** A capability already demonstrated reduces the time, cost, and risk of the next pathway.

Accountability matters more than a qualification declaration. Every item of evidence has a provenance, scope of applicability, assessor, and review procedure. **A bounded and contestable model matters more than total observability.** Evidence of capability is portable; a permanent history of errors and drafts is not.

Continuity includes adaptation, development, and exploration. For the working adult, learning becomes a third job—after paid employment and family. Time, cost, and support must therefore be designed as part of a professional and social system, not as a private hobby after the working day.

42. Working Definition

Quantum learning is an event-driven system, implemented through digital twins, for managing minimal, verified transitions in human capability.

Every transition:

- is triggered by an event
- is connected to a specific gap
- is executed as a distinct process
- includes human action
- records attempts and assistance
- ends with assessment
- creates evidence
- changes the state of the digital twin
- affects authorization for subsequent actions
- restructures the individual learning graph.

Quantum learning is not a complete theory of education and does not claim to provide a complete model of the human being. It is an architecture for managing demonstrable transitions in domains where the action, context, and sufficient evidence can be substantively defined. In development and exploration, the system performs a supportive rather than exhaustive function. In applied domains, basic target states may be assembled into a professional trivium—a compact list of tasks that a qualified specialist must be able to perform independently. Quanta construct the individual pathway to those tasks, and evidence shows that the necessary minimum has genuinely been attained. Quantum learning does not refer merely to:

- microlearning
- an adaptive test
- an AI tutor
- a personalized course
- a content library
- a digital diploma
- an updated LMS.

It is an architecture in which all participants and consequential objects in the educational process are represented by digital twins, while human development is executed as an individual graph of verified transitions.

43. Manifesto

Unit and Outcome

We demand that education measure changes in capability, not hours, credits, and the volume of content consumed. We demand that the mandatory common program be replaced by an individual graph beginning with the person's current state and the requirements of a specific task or goal.

We demand that every professionally consequential transition end in evidence and, where necessary, a transparent authorization decision.

Actors and Execution

We demand that all consequential participants and objects in the process be represented by digital twins in an executable graph of states, events, and accountability. We demand that the employer become an active participant by describing real tasks and risks, providing context, participating in assessment, and returning the actual result.

We demand that AI be used to diagnose and adapt transitions, but never be granted unilateral authority to assign an irreversible status to a person.

Economics and Accountability

We demand that the student pay for the pathway actually required, and that recognized capabilities reduce its duration and price. We demand that the chain of the qualification claim be traceable—from the task model and diagnosis to the quantum, assessment, evidence, and authorization.

We demand that the university become an operator of the continuous development of people and companies; that it design learning for the working adult as a third job compatible with paid employment and family; and that teacher expertise scale through reusable quanta, simulations, and work on complex cases.

Rights and Openness

We demand that the digital model be bounded, transparent, portable, and contestable, rather than becoming a universal rating of the person. We demand separation of the private learning history from portable evidence, data minimization, and defined retention periods for learning traces.

We demand the preservation of autonomy in development and exploration: a known gap and current employer demand do not exhaust the purposes of education.

44. The First Move: One Task, One Person, One Quantum

Quantum learning does not require beginning with the reform of an entire university, industry, or company. The first operating circuit is built around one real task, one person, and one demonstrable transition. The employer describes the required result, work context, critical errors, and admissible risk. The university or another provider records the initial state, designs one quantum, and defines the assessment procedure. The person follows an individual pathway, while the teacher, mentor, expert, and AI enter only to the extent required for that transition. The circuit is closed not when the learning material has been completed, but when evidence has been created, an authorization decision made, and the actual work result returned to the digital twins of all participants. This minimum scale makes it possible to test the architecture, economics, access rights, and allocation of accountability before the system is expanded.

MINIMUM OPERATING CIRCUIT

ONE TASK → ONE PERSON → ONE QUANTUM → ONE ITEM OF EVIDENCE → ONE AUTHORIZATION

Conclusion

The principal crisis of modern education lies not only in its duration. It lies in the mismatch between the form of the educational process and the speed at which activity changes. The system continues to transmit knowledge in large packages, while the need for new capabilities arises continuously, locally, and in response to events. Shortening a course, moving it online, or adding AI-generated content does not change the primary unit. At the labor-market level, the cost of the legacy unit appears as structural unemployment: vacancies and people coexist, but no demonstrable, safe, and economically admissible transition has been constructed between their graphs.

Quantum learning becomes an adaptation layer for the labor market, but it does not reduce the person to a set of functions or subordinate all education to current business demand. The digital model remains bounded, learning errors remain protected, and the university preserves a circuit for long-horizon development and open exploration. The primary unit must change. In place of the course—a verified transition. In place of the common program—an individual graph.

In place of completion and certification—action and evidence. In place of payment for the entire package—payment for the pathway actually required, with recognition of what has already been demonstrated. In place of a one-time sale of a degree—continuous stewardship of the development of the person and the company. In place of diffuse accountability—a traceable chain from the task and quantum to evidence and authorization. **A course measures content. A quantum measures change in the person.**

Education must cease to be a period of life. It must become a continuous system of precise, timely, and verifiable transitions.

This is quantum learning.

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